

Elementary Problems
Angle Chasing
Solutions

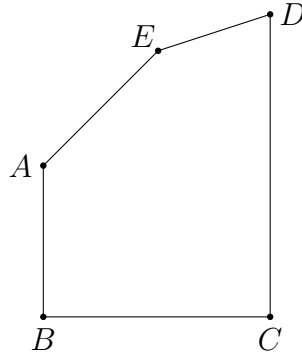
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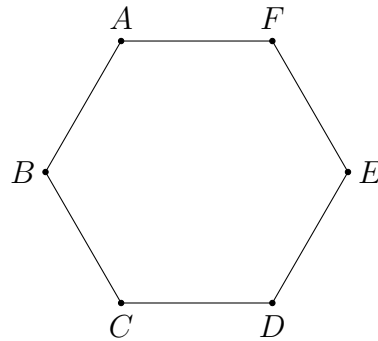
Basic Work

1. The sum of interior angles of a pentagon is $(5 - 2) \times 180^\circ = 540^\circ$. Thus,

$$\angle E = 540^\circ - 135^\circ - 90^\circ - 90^\circ - 72^\circ = 153^\circ.$$



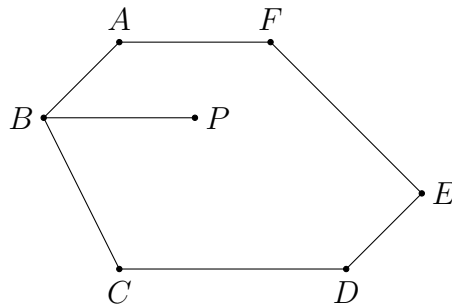
2. The sum of interior angles of a hexagon is $(6 - 2) \times 180^\circ = 720^\circ$. Thus, each interior angle is $\frac{720^\circ}{6} = 120^\circ$.



3. Let P be a point inside the hexagon for which $BP \parallel AF$. Then

$$\angle A + \angle B + \angle C = \angle A + \angle PBA + \angle CBP + \angle C = 180^\circ + 180^\circ = 360^\circ$$

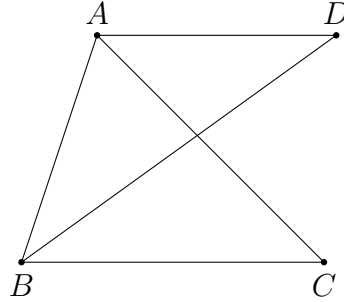
from the parallel lines.



4. We have

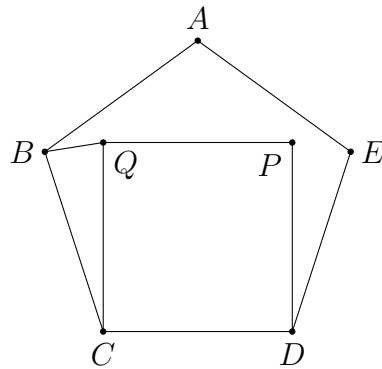
$$\angle ADB = \angle CBD = \angle DBA.$$

This implies $AB = AD$.



5. Note that $CQ = CD = CB$. Hence,

$$\angle BQC = \frac{180^\circ - \angle QCB}{2} = \frac{180^\circ - \angle DCB + \angle DCQ}{2} = \frac{180^\circ - 108^\circ + 90^\circ}{2} = 81^\circ.$$

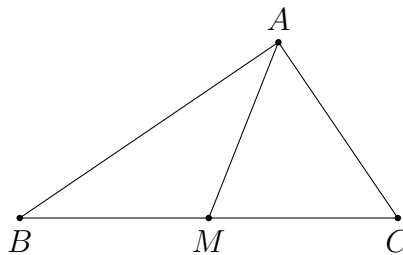


6. If $MA = MB$, then we have $MA = MB = MC$. Let $\angle MBA = \angle BAM = \alpha$ and $\angle ACM = \angle MAC = \beta$. By considering the angle sum of triangle, we have

$$\alpha + \alpha + \beta + \beta = 180^\circ.$$

This implies $\alpha + \beta = 90^\circ$. Hence,

$$\angle BAC = \angle BAM + \angle MAC = \alpha + \beta = 90^\circ.$$



If $\angle BAC = 90^\circ$, let M' be the point on BC such that $M'A = M'B$. Then

$$\angle M'AC = 90^\circ - \angle BAM' = 90^\circ - \angle M'BA = \angle ACM'.$$

This implies $M'A = M'C$ and hence $M' = M$. Thus, $MA = MB$.

7. It suffices to show N lies on AF and BC . From the concyclic points and the property of squares, we have

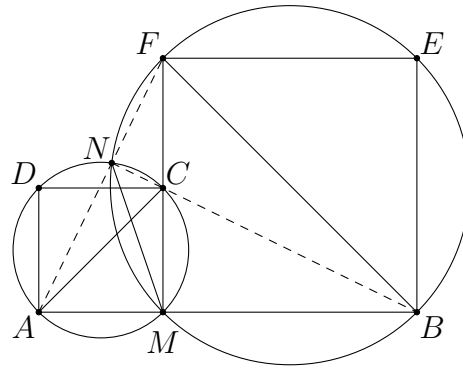
$$\angle ANM + \angle MNF = \angle ACM + (180^\circ - \angle FBM) = 45^\circ + (180^\circ - 45^\circ) = 180^\circ.$$

Thus, A, N, F are collinear.

Similarly, we have

$$\angle MNC = \angle MAC = 45^\circ = \angle MFB = \angle MNB.$$

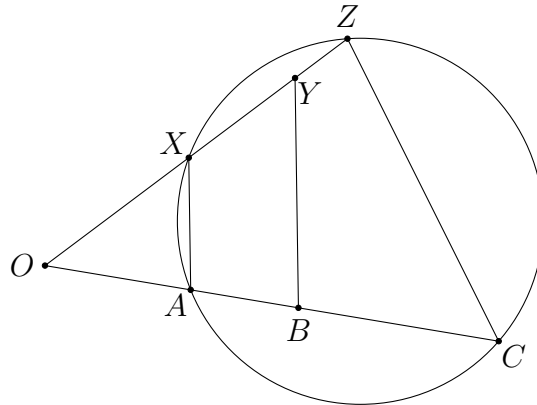
Thus, N, C, B are collinear.



8. We have

$$\angle YBO = \angle XAO = \angle XZC = \angle YZC.$$

This implies B, C, Z, Y are concyclic.



9. We have

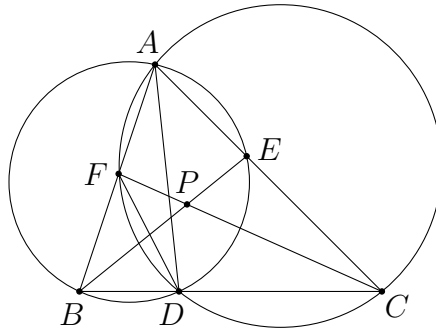
$$\angle DBP = \angle DBE = \angle DAE = \angle DAC = \angle DFC = \angle DFP.$$

This shows B, D, P, F are concyclic. Similarly, C, D, P, E are concyclic.

Next, we have

$$\angle AEP + \angle PFA = \angle AEB + \angle CFA = \angle ADB + \angle CDA = 180^\circ.$$

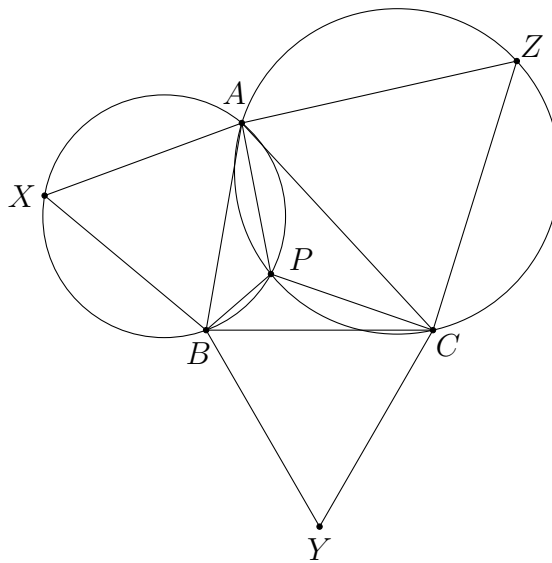
This shows A, F, P, E are concyclic.



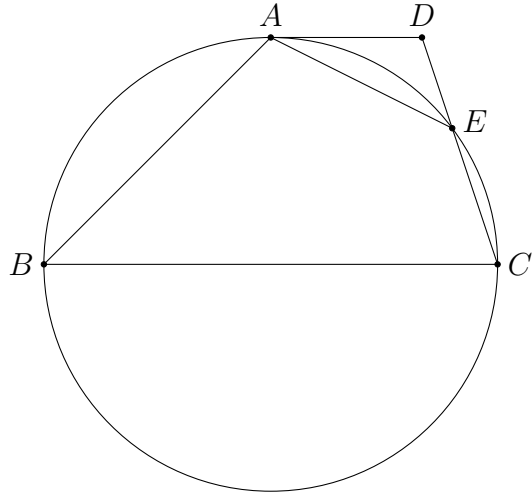
10. Let (XBA) and (ZAC) meet again at P . Then

$$\begin{aligned} \angle BPC &= 360^\circ - \angle APB - \angle CPA \\ &= 360^\circ - (180^\circ - \angle BXA) - (180^\circ - \angle AZC) \\ &= 360^\circ - (180^\circ - 60^\circ) - (180^\circ - 60^\circ) \\ &= 120^\circ = 180^\circ - \angle CYB. \end{aligned}$$

Hence, P lies on (YCB) . The result then follows.



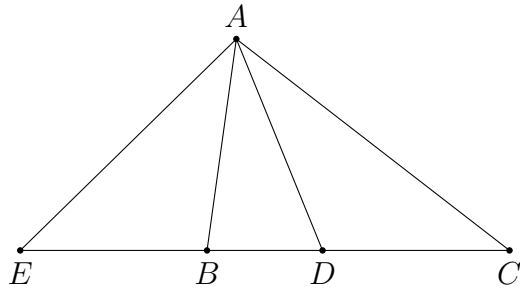
11. Since A, B, C, E are concyclic, we have $\angle BAE = 180^\circ - \angle ECB = \angle ADE$. Therefore, AB is tangent to (DAE) .



12. Since E lies on the perpendicular bisector of AD , we have $\angle EAD = \angle ADE$. Hence,

$$\angle EAB = \angle EAD - \angle BAD = \angle ADE - \angle DAC = \angle ACB.$$

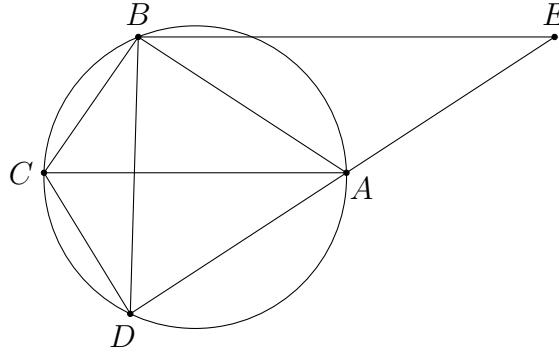
This implies EA is tangent to (ABC) .



13. We have

$$\angle BEA = \angle CAD = \angle CBD = \angle BDC = \angle BAC = \angle ABE.$$

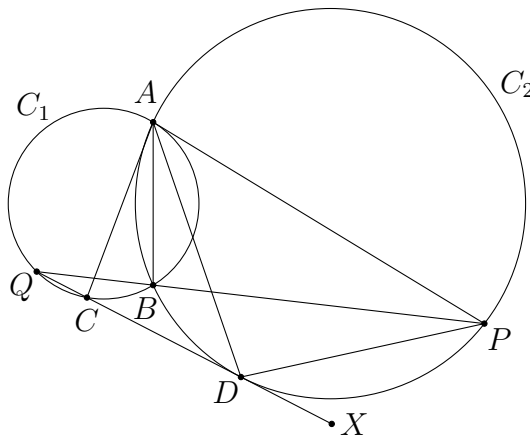
This implies $AB = AE$.



14. Let X be a point on the ray CD beyond D . We have

$$\angle CAD = \angle CAB + \angle BAD = \angle CQB + \angle BPD = \angle XDP = \angle DAP.$$

Thus, AD bisects $\angle CAP$.



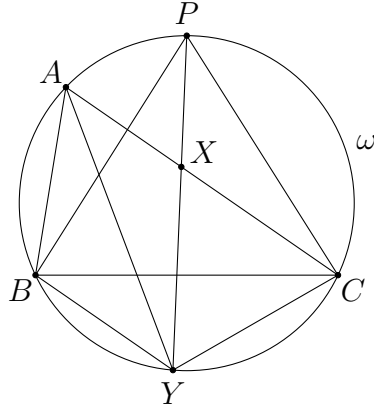
15. Since $ABYC$ is cyclic and $AB = CY$, it is an isosceles trapezoid. From this, we obtain $\angle ACB = \angle YAC$. Hence, we have

$$\angle PCB = \angle PCA + \angle ACB = \angle PYA + \angle YAC = \angle YXC.$$

Using $CX = CY$, we get

$$\angle PCB = \angle YXC = \angle CYX = \angle CYP = \angle CBP.$$

Hence, $PB = PC$.



16. Firstly, we have

$$\angle BAP = \angle CAP - \angle CAB = \angle DBF - \angle CEB.$$

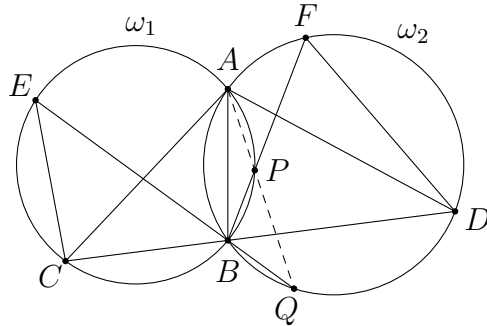
Secondly, we have

$$\angle BAQ = \angle BAD - \angle QAD = \angle BFD - \angle QBD = \angle BFD - \angle EBC.$$

Note that $\angle DBF = \angle BFD$ and $\angle CEB = \angle EBC$ since $DB = DF$ and $CB = CE$. Therefore, we have

$$\angle BAP = \angle BAQ.$$

This shows A, P, Q are collinear.



17. Since $AC = AE$, we have

$$\angle BEF = \angle CEA = \angle ACE = \angle ACB = \angle EFB.$$

This implies $BE = BF$.

Next, we have

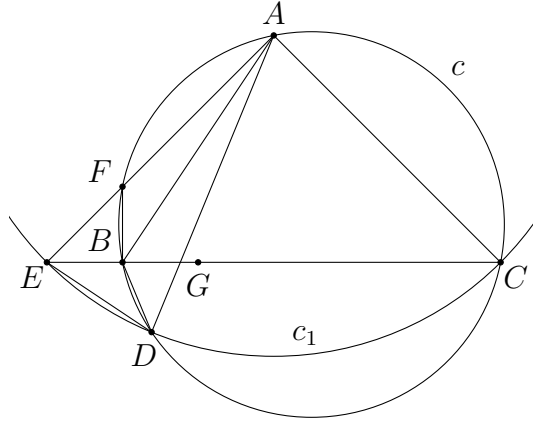
$$\angle ADB = \angle ACB = \angle ACE = \angle CEA = \angle BEA.$$

This yields

$$\angle BDE = \angle ADE - \angle ADB = \angle DEA - \angle BEA = \angle DEB,$$

and hence $BD = BE$.

Therefore, F, E, D, G lie on a circle with centre B . Thus, they are concyclic.

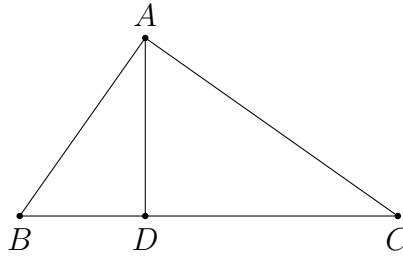


Finding Similar Triangles and Concyclic Points

18. Since $\angle ADB = 90^\circ = \angle CAB$ and $\angle DBA = \angle ABC$, we have $\triangle ABD \sim \triangle CBA$.

Hence, we get $\frac{AB}{CB} = \frac{BD}{BA}$, i.e. $BD \times BC = BA^2$.

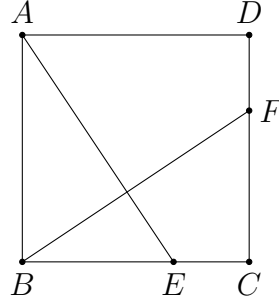
Similarly, we have $\triangle CAD \sim \triangle CBA$. So $\triangle ABD \sim \triangle CAD$. This implies $\frac{AD}{CD} = \frac{BD}{AD}$, i.e. $BD \times CD = AD^2$.



19. From $AB = BC$, $BE = CF$ and $\angle ABE = 90^\circ = \angle BCF$, we have $\triangle ABE \cong \triangle BCF$. Hence, we have $\angle AEB = \angle BFC$. This yields

$$\angle AEB + \angle CBF = \angle BFC + \angle CBF = 90^\circ.$$

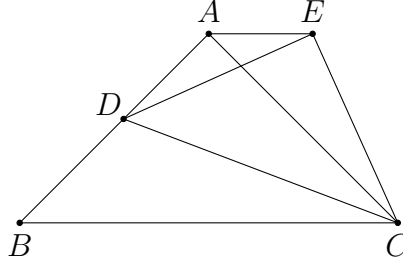
Thus, $AE \perp BF$.



20. Since $\angle DAC = 90^\circ = \angle DEC$, points A, D, C, E are concyclic. Hence, we have

$$\angle CAE = \angle CDE = 45^\circ = \angle ACB.$$

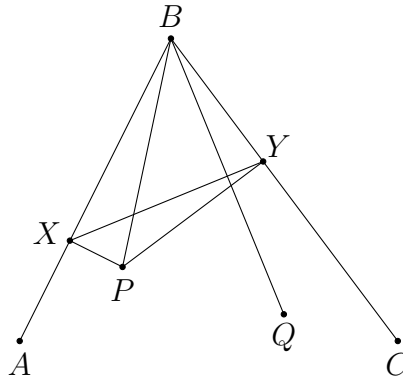
Thus, $AE \parallel BC$.



21. Since $\angle PXB = \angle BYP = 90^\circ$, points B, X, P, Y are concyclic. From this, we obtain $\angle XYP = \angle XBP$. It follows that

$$\angle QBY + \angle BYX = \angle XBP + \angle BYX = \angle XYP + \angle BYX = 90^\circ.$$

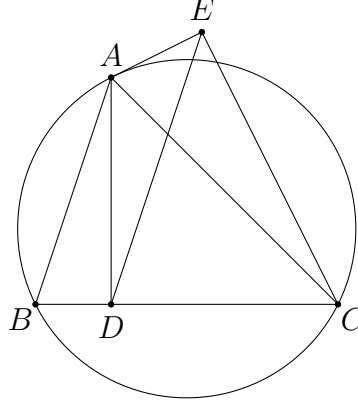
Hence, $XY \perp BQ$.



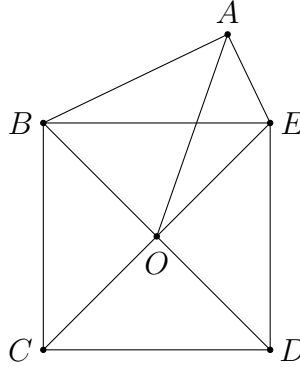
22. We have

$$\angle CAE = \angle CBA = \angle CDE.$$

This implies A, D, C, E are concyclic. Thus, $\angle AEC = 180^\circ - \angle CDA = 90^\circ$.



23. Note that $\triangle BOE$ is a right-angled and isosceles triangle with $\angle EOB = 90^\circ$. From $\angle EOB = \angle BAE = 90^\circ$, points A, B, O, E are concyclic. Since $OB = OE$, this implies $\angle BAO = \angle OAE$.



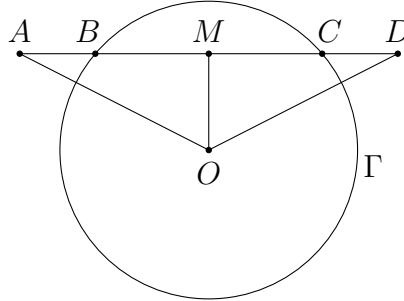
24. Firstly, suppose $OA = OD$. Let M be the midpoint of AD . Then $MA = MD$. Together with the common side OM , we get $\triangle AMO \cong \triangle DMO$. Hence, $\angle AMO = \angle DMO$. As their sum is 180° , each of them is 90° . Since $OM \perp BC$ and O is the centre of Γ , M is the midpoint of BC . This yields

$$AB = AM - BM = DM - CM = CD.$$

Conversely, suppose $AB = CD$. Let M be the midpoint of BC . Then

$$AM = AB + BM = CD + CM = DM.$$

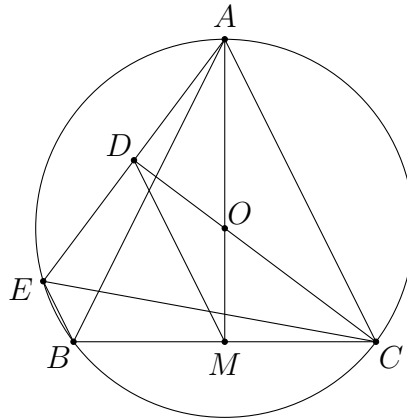
Also, since $OM \perp BC$, we have $\angle AMO = \angle DMO = 90^\circ$. Together with the common side OM , we get $\triangle AMO \cong \triangle DMO$. Thus, $OA = OD$.



25. As $\angle CMA = 90^\circ = \angle CDA$, points A, D, M, C are concyclic. Thus,

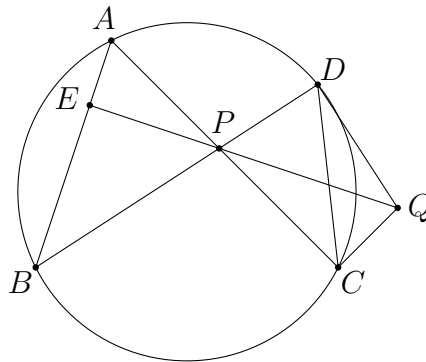
$$\angle MDA = 180^\circ - \angle ACM = 180^\circ - \angle ACB = \angle BEA.$$

This implies $EB \parallel DM$. As M is the midpoint of BC , the line DM bisects EC .



26. Let AB meet PQ at E . As $\angle PDQ = \angle QCP = 90^\circ$, points C, Q, D, P are concyclic. Thus,

$$\angle APE + \angle EAP = \angle CPQ + \angle BAP = \angle CDQ + \angle BDC = 90^\circ.$$

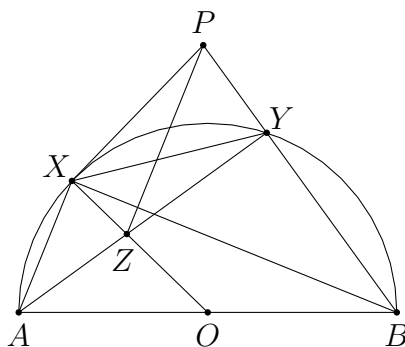


This implies $AB \perp PQ$.

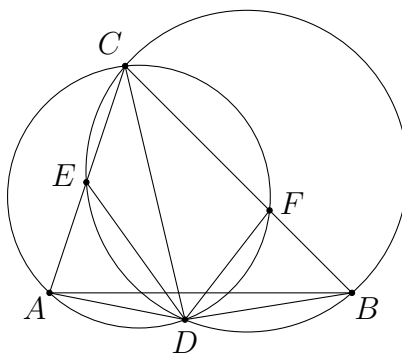
27. Since PX is tangent to the semicircle, we have $\angle OXP = 90^\circ$. Also, as AB is the diameter of the semicircle, we have $\angle AYB = 90^\circ$. These show P, X, Z, Y are concyclic. Then

$$\angle XPZ = \angle XYZ = \angle XBA = 90^\circ - \angle BAX = 90^\circ - \angle BXP.$$

This yields $PZ \perp BX$.



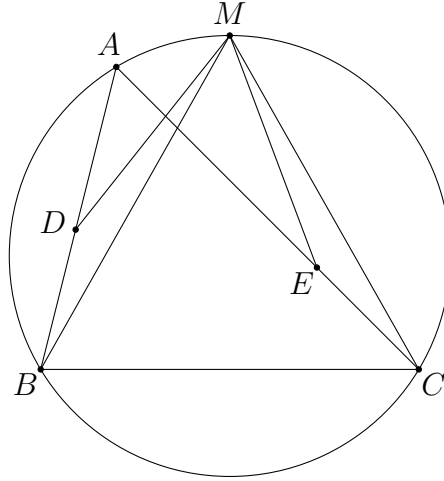
28. From the given conditions, we have $\angle AED = \angle FBD$, $\angle DAE = \angle DFB$ and $AE = FB$. These yield $\triangle ADE \cong \triangle FDB$. Hence, we get $DE = DB$. As C, E, D, B are concyclic, this implies CD bisects $\angle ECB$. Thus, CD bisects $\angle ACB$.



29. Since M lies on the perpendicular bisector of BC , we have $MB = MC$. Also, we have $BD = CE$ and $\angle MBD = \angle MCE$. Therefore, $\triangle DBM \cong \triangle ECM$. This yields $\angle BDM = \angle CEM$. Hence,

$$\angle MDA = 180^\circ - \angle BDM = 180^\circ - \angle CEM = \angle MEA.$$

Thus, A, D, E, M are concyclic.

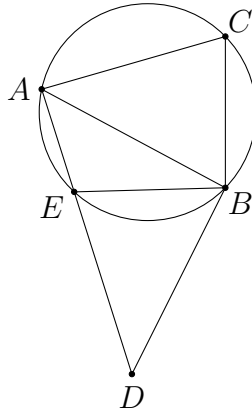


30. Firstly, assume $AB = BD$. We have

$$\angle BDE = \angle BDA = \angle DAB = \angle BAC.$$

Also, we have $\angle DEB = \angle ACB$. These imply $\triangle BED \sim \triangle BCA$. As AB bisects $\angle EAC$, we have $BE = BC$. Hence, $\triangle BED \cong \triangle BCA$. It follows that $ED = CA$.

Next, assume $ED = CA$. Again, we have $BE = BC$ and $\angle DEB = \angle ACB$. Hence, $\triangle BED \cong \triangle BCA$. It follows that $BD = BA$.

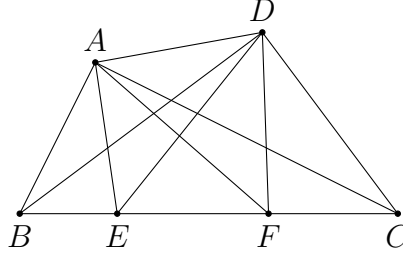


31. From $\angle EAF = \angle EDF$, we know that A, E, F, D are concyclic. Therefore, we have $\angle ADF = \angle AEB$. It follows that

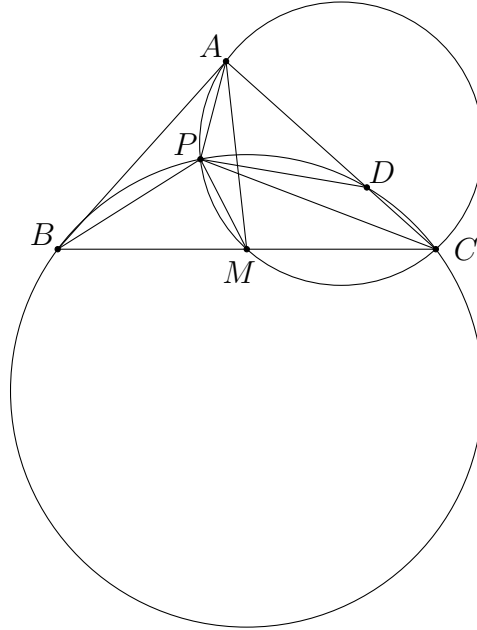
$$\angle ADC + \angle CBA = \angle ADF + \angle FDC + \angle CBA = \angle AEB + \angle BAE + \angle EBA = 180^\circ.$$

This shows A, B, C, D are concyclic. Thus,

$$\angle FAC = \angle BAC - \angle BAE - \angle EAF = \angle BDC - \angle FDC - \angle EDF = \angle BDE.$$



32. Note that $MB = MA = MC$. Thus, we have $MB = AD$. From the concyclic points, we have $\angle PMB = \angle PAD$ and $\angle MBP = \angle ADP$. These imply $\triangle PBM \cong \triangle PDA$. It follows that $PM = PA$. As A, P, M, C are concyclic, we find that CP bisects $\angle ACB$.



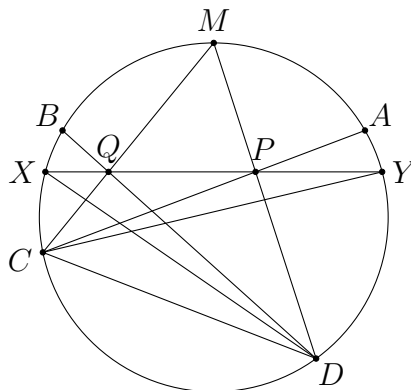
33. As M is the midpoint of the arc AB , we have $\angle ACM = \angle MDB$. Thus, C, D, P, Q are concyclic. This implies $\angle QPC = \angle QDC$. Hence,

$$\angle BDX = \angle BDC - \angle XDC = \angle XPC - \angle XYC = \angle YCA.$$

Together with $\angle MDB = \angle ACM$, we find that

$$\angle MDX = \angle MDB + \angle BDY = \angle ACM + \angle YCA = \angle YCM.$$

Thus, $MX = MY$.

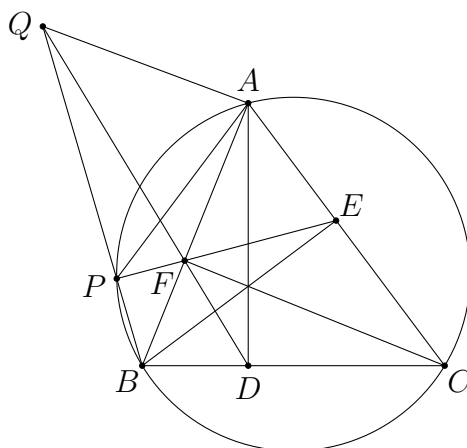


34. There are two configurations to consider. We first deal with the figure below.

Since A, P, B, C are concyclic, we have $\angle APQ = \angle ACB$. Recall that A, F, D, C are concyclic since $\angle CFA = 90^\circ = \angle CDA$. Thus, we have $\angle ACB = \angle BFD = \angle AFQ$. We have found that

$$\angle APQ = \angle AFQ.$$

This implies A, Q, P, F are concyclic. Therefore, we have $\angle PQA = \angle EFA$. Again, as E, F, B, C are concyclic, we have $\angle EFA = \angle ACB = \angle APQ$. Hence, $\angle PQA = \angle APQ$. It follows that $AP = AQ$.



Next, we deal with the configuration below. The proof is very similar to the former case.

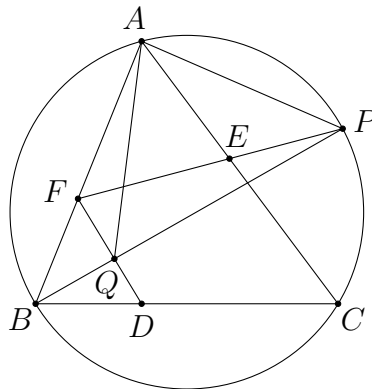
We have

$$\angle APQ = \angle ACB = \angle BFQ.$$

This implies A, F, Q, P are concyclic. Therefore, we have

$$\angle PQA = \angle PFA = \angle ACB = \angle APQ.$$

It follows that $AP = AQ$.



35. We have

$$90^\circ = \angle XBY = \angle XBA + \angle ABY = \angle XX'A + \frac{1}{2}\angle AO_1Y.$$

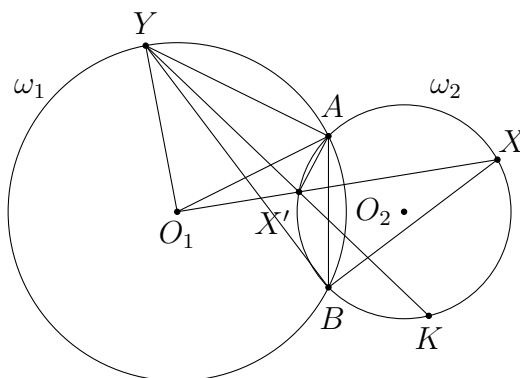
Note that $O_1A = O_1Y$. Thus,

$$\angle O_1YA = \frac{180^\circ - \angle AO_1Y}{2} = 90^\circ - \frac{1}{2}\angle AO_1Y = \angle XX'A.$$

This shows A, Y, O_1, X' are concyclic. It follows that

$$\angle KX'X = \angle YX'O_1 = \angle YAO_1 = \angle O_1YA = \angle XX'A.$$

Hence, X is the midpoint of the arc AK .



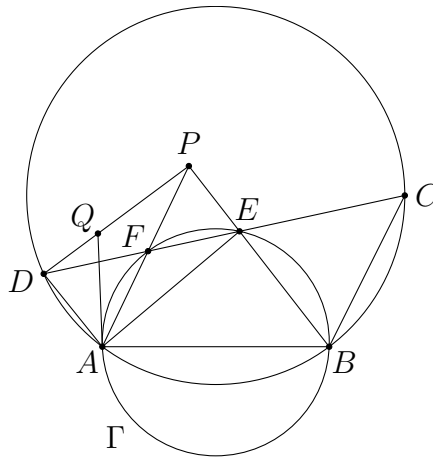
36. Let DP meet the tangent at A to Γ at Q . We need to prove $QA = QP$. From

$$\begin{aligned}\angle ADE &= 180^\circ - \angle CBA = 180^\circ - \angle CBE - \angle EBA \\ &= 180^\circ - \angle BEC - \angle EFP = 180^\circ - \angle PEF - \angle EFP = \angle APE,\end{aligned}$$

we find that D, A, E, P are concyclic. It follows that

$$\angle QPA = \angle DPA = \angle DEA = \angle FEA = \angle FAQ = \angle PAQ.$$

Thus, Q lies on the perpendicular bisector of AP as desired.

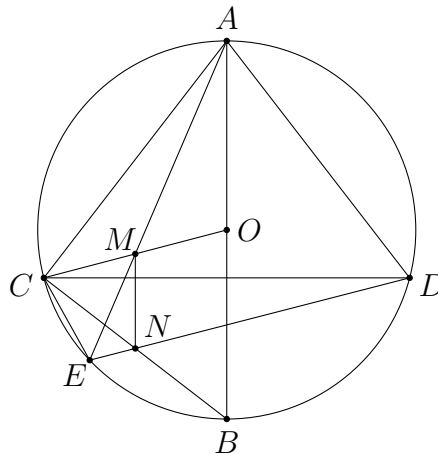


37. Let AE meet CO at M and let BC meet DE at N .

By symmetry, A is the midpoint of the arc CD . Therefore, we have $\angle DEA = \angle ABC$. Since $OB = OC$, we have

$$\angle NCM = \angle BCO = \angle OBC = \angle ABC = \angle DEA = \angle NEM.$$

This implies C, E, N, M are concyclic.



From this, we obtain

$$\begin{aligned}\angle CMN &= 180^\circ - \angle NEC = 180^\circ - \angle DEC \\ &= \angle CAD = 2\angle CAB = \angle COB.\end{aligned}$$

This implies $MN \parallel OB$. Since M is the midpoint of CO , N is the midpoint of CB . So DE bisects BC .

38. Firstly, we have

$$\angle CMP = B + \angle BAM = B + \frac{A}{2}$$

and

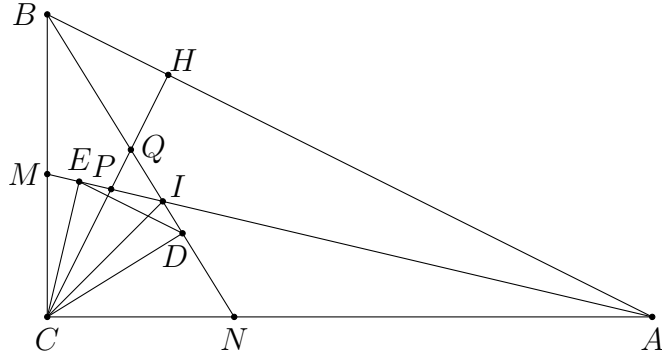
$$\angle MPC = \angle MAC + \angle ACH = \frac{A}{2} + (90^\circ - A) = \frac{A}{2} + B = \angle CMP.$$

This shows $CM = CP$. Let E be the midpoint of MP . Then $CE \perp MP$.

Similarly, let D be the midpoint of NQ . Then $CD \perp NQ$. Let I be the incentre of $\triangle ABC$, i.e. I is the intersection of AM and BN . From $\angle IDC = \angle CEI = 90^\circ$, points C, D, I, E are concyclic. Then we have

$$\begin{aligned}\angle IDE &= \angle ICE = \angle ICB - \angle ECM = \frac{C}{2} - \frac{1}{2}\angle PCM \\ &= 45^\circ - \frac{1}{2}(90^\circ - B) = \frac{B}{2} = \angle IBA.\end{aligned}$$

Therefore, we get $DE \parallel AB$.



39. By the midpoint theorem, we have $DE \parallel BC$. As $\angle BAF = \angle DEC$, we get

$$\angle FCD = 180^\circ - \angle BAF = 180^\circ - \angle DEC = \angle ACB.$$

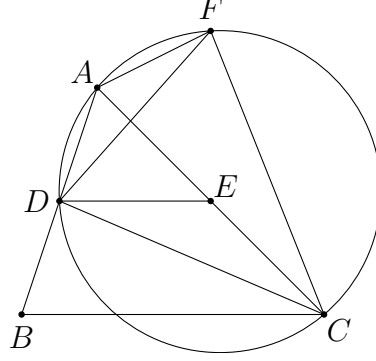
Thus,

$$\angle FCA = \angle FCD - \angle ACD = \angle ACB - \angle ACD = \angle DCB.$$

Together with $\angle AFC = \angle BDC$, we find that $\triangle FAC \sim \triangle DBC$. It follows that

$$\frac{FC}{FA} = \frac{DC}{DB} = \frac{DC}{DA}$$

as D is the midpoint of AB . The result then follows.



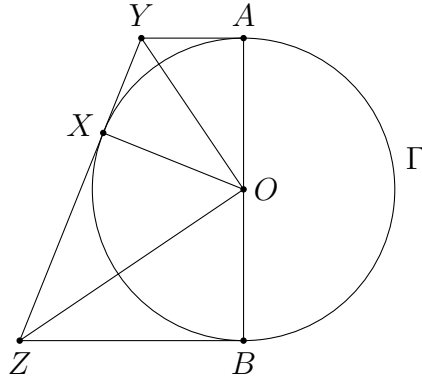
40. Let O be the centre of Γ . Then $\triangle YAO \cong \triangle YXO$ and $\triangle ZBO \cong \triangle ZXO$. Let $\angle AOY = \angle XOY = \alpha$ and $\angle BOZ = \angle XOZ = \beta$. Then

$$\alpha + \alpha + \beta + \beta = 180^\circ,$$

which implies $\angle YOZ = \alpha + \beta = 90^\circ$. As $OX \perp YZ$, we have $\triangle OXY \sim \triangle ZXO$. Therefore, we obtain $\triangle OAY \sim \triangle ZBO$. This yields

$$\frac{OA}{ZB} = \frac{AY}{BO}.$$

Thus, $AY \cdot BZ = OA \cdot OB$ is a constant since OA and OB are radii of Γ .



41. Due to symmetry, O must be the midpoint of BC . Suppose

$$PB \times CQ = \left(\frac{BC}{2}\right)^2 = BO \times OC.$$

$$\frac{PB}{BO} = \frac{OC}{CQ}.$$
$$\frac{PB}{BO} = \frac{PO}{OQ}.$$
$$\angle OBP = 180^\circ - \angle BPO - \angle POB = 180^\circ - \angle COQ - \angle POB = \angle QOP.$$
$$P'B = \frac{1}{CQ} \left(\frac{BC}{2} \right)^2.$$

42. As $\angle APB = 90^\circ = \angle BQC$ and

we have $\triangle ABP \sim \triangle BCQ$. This yields

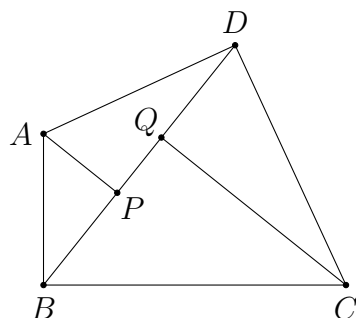
$$\frac{AP}{BP} = \frac{BQ}{CQ}. \quad (1)$$

Similarly, we have $\triangle ADP \sim \triangle DCQ$ and hence

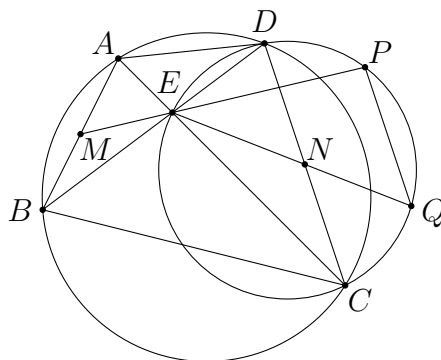
$$\frac{AP}{DP} = \frac{DQ}{CQ}. \quad (2)$$

Combining (1) and (2), we get $\frac{DP}{BP} = \frac{BQ}{DQ}$. WLOG assume B, P, Q, D lie in that order. Then

$$\begin{aligned} & \frac{DP}{BP} = \frac{BQ}{DQ} \\ \Rightarrow & \frac{DP}{BP} + 1 = \frac{BQ}{DQ} + 1 \\ \Rightarrow & \frac{BD}{BP} = \frac{BD}{DQ} \\ \Rightarrow & BP = DQ. \end{aligned}$$



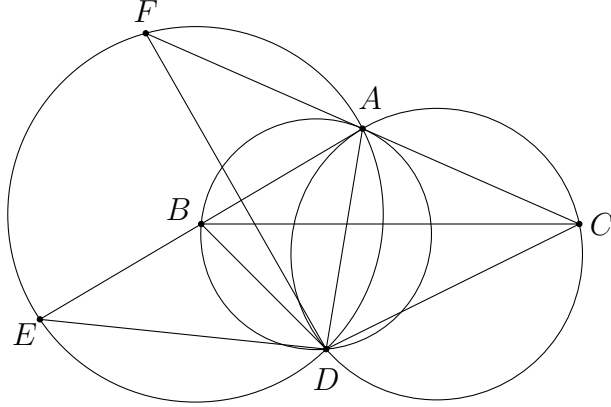
43. Note that $\triangle ABE \sim \triangle DCE$. As M and N are the midpoints of the corresponding sides AB and DC , we have $\triangle EMB \sim \triangle ENC$. This implies $\angle MEB = \angle NEC$. Hence, we have $\angle PED = \angle CEQ$. This shows the arcs DP and CQ have equal length. Thus, $DCQP$ is an isosceles trapezoid. In particular, $CD \parallel QP$.



44. Firstly, from $\angle DBA = \angle DAC$ and $\angle BAD = \angle ACD$, we have $\triangle DBA \sim \triangle DAC$.

Secondly, from $\angle DEA = \angle DFC$ and $\angle EAD = \angle FCD$, we have $\triangle EAD \sim \triangle FCD$.

These show $DEBA$ and $DFAC$ are similar. Since B is the midpoint of EA , the corresponding point A must be the midpoint of FC . This yields $AF = AC$.



45. Let P be the foot of the altitude from B . Since $\angle BPC = 90^\circ = \angle AED$ and

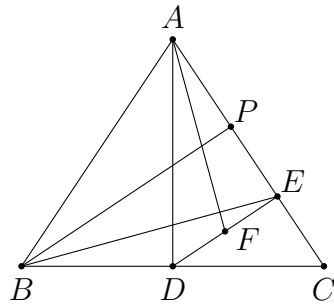
$$\angle PCB = 90^\circ - \angle CDE = \angle EDA,$$

we have $\triangle BCP \sim \triangle ADE$.

Next, note that $BP \parallel DE$ since both are perpendicular to AC . As D is the midpoint of CB , E is the midpoint of CP . Hence, the midpoint E of CP corresponds to the midpoint F of DE under the above similarity. This gives $\angle FAE = \angle EBP$. Hence,

$$\angle FAE + \angle AEB = \angle EBP + \angle AEB = 90^\circ$$

as $\angle BPE = 90^\circ$. This implies $AF \perp BE$.



46. From $\angle BAP = \angle ACQ$ and $\angle PBA = \angle QAC$, we have $\triangle BPA \sim \triangle AQC$. This implies

$$\frac{BP}{AQ} = \frac{PA}{QC}.$$

Using $AQ = NQ$ and $PA = PM$, we get

$$\frac{BP}{NQ} = \frac{PM}{QC}.$$

Together with

$$\angle BPM = 180^\circ - \angle APB = 180^\circ - \angle CQA = \angle NQC,$$

we obtain $\triangle PBM \sim \triangle QNC$. This yields $\angle PMB = \angle QCN$. Let D be the intersection of BM and CN . Then

$$\begin{aligned} \angle CDB &= 180^\circ - \angle BCD - \angle DBC = 180^\circ - \angle PMB - \angle MBP \\ &= \angle BPM = \angle PBA + \angle BAP = \angle CBA + \angle ACB \\ &= 180^\circ - \angle BAC. \end{aligned}$$

This shows D lies on (ABC) .

